

IADNet: A neural network for discovering particle interactions from trajectory data [★]

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ABSTRACT

Deciphering interparticle interactions remains fundamental to unraveling complex systems across physics, chemistry, and biological sciences. Conventional approaches often suffer from prior knowledge reliance or insufficient interpretability in characterizing interactions. To overcome these limitations, this study proposes the Interaction Discovery Neural Network (IADNet), which integrates graph structure learning with physics-constrained strategies to directly reconstruct interaction graph topologies from particle trajectories while establishing explicit mapping relationships with actual physical system configurations. To validate its universality, we tested IADNet across four different systems spanning classical mechanics, molecular chemistry, biological models, and lattice systems. Experimental results demonstrate that IADNet can accurately reconstruct spring connection topologies and estimate elastic coefficient ratios, effectively identify chemical bonds in molecules, reveal hidden long-range interactions in DNA models and restore base sequence arrangements, and detect anomalous patterns in lattice systems. These findings underscore IADNet's potential in multiparticle system research.

1. Introduction

Hamiltonian systems involving multi-particle interactions underpin a vast range of phenomena in condensed matter physics [1], materials science [2], chemistry [3], biology [4], biochemistry [5], and medicine [6]. A prerequisite for quantitative understanding in all these domains is an accurate description of the *interaction network*: which particles interact, and with what strength. Microscopic knowledge of this network delivers practical benefits. In materials science, identifying particle interactions is crucial for understanding microscopic defects, such as vacancies, substitutions, and dislocations. These interactions form the theoretical underpinnings for the design and optimization of materials with desired properties. In chemistry, identifying chemical bonds allows functional groups to be recognized, enabling the reliable prediction of reactivity. In pharmaceutical science, understanding intermolecular interactions and the crystalline structure of drug solids is key to controlling critical parameters, such as drug solubility and bioavailability, thus serving as a vital basis for innovative drug formulation strategies.

However, in many practical scenarios, due to limitations in experimental techniques or the inherent complexity of particle interactions, it remains a challenge to accurately observe and identify these interac-

tions. Especially when interactions are localized to nearest or specific-range neighbors, the problem of identifying particle interactions becomes even more intricate. This challenge raises a fundamental question: *Can we recover the latent interaction graph directly from particle trajectories, without any a priori structural information?* Fig. 1 visualizes the problem. We record only kinematic data (panel b), whereas the true couplings (panel a) are hidden; the task is therefore to reconstruct an explicit interaction graph (panel c) solely from the observed motion.

The advent of artificial neural networks has revolutionized the field of complex data analysis. Nevertheless, conventional neural network techniques face significant difficulties in modeling multiparticle interactions due to issues such as fully-connected architectures ignoring relational structure, suffering from input-order sensitivity, and demanding prohibitive data volumes. Graph neural networks (GNNs) alleviate many of these issues by representing particles as nodes and interactions as edges, propagating information via message passing with inductive biases that respect permutation symmetry. However, nearly all existing GNN frameworks *assume* the graph is known. They either take the interactions' existence as given and learn interaction strengths from trajectories [7–9], or they rely on known local structures for

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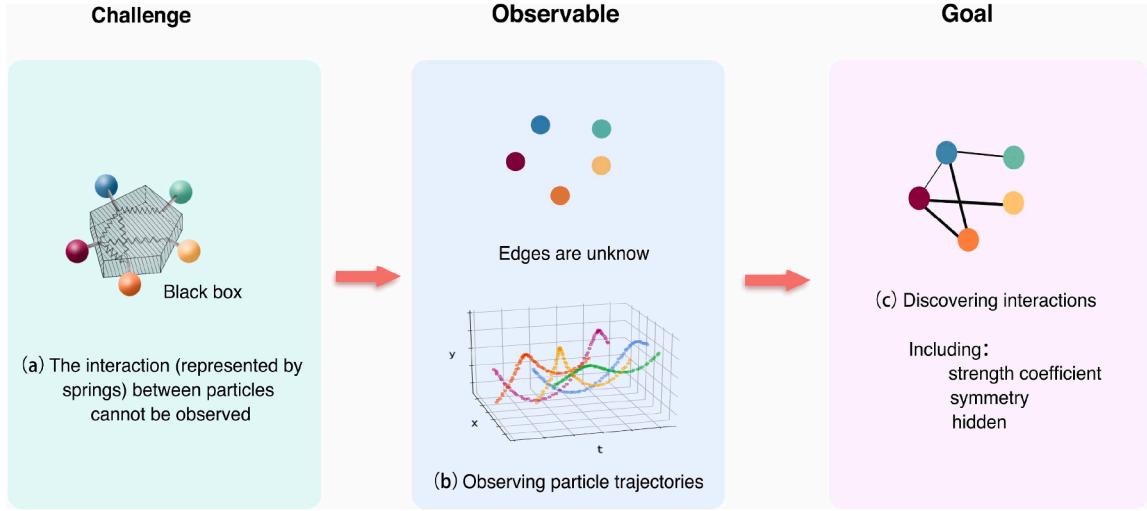


Fig. 1. Inferring hidden interactions from motion. (a) Ground-truth but unobserved spring connections between particles. (b) Measured trajectories, the only data available to the learner. (c) Desired output: an explicit interaction graph in which edges and weights faithfully reproduce the hidden couplings.

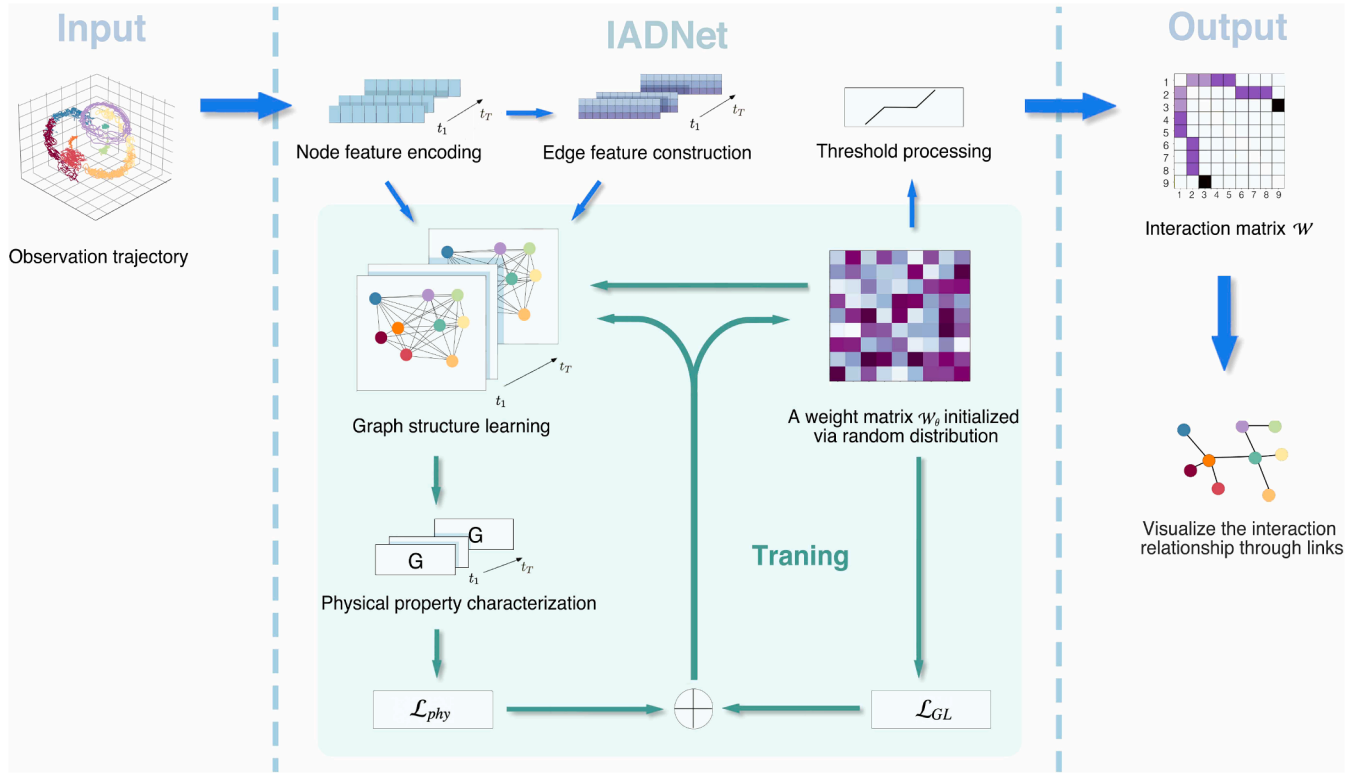


Fig. 2. Workflow of the interaction discovery neural network (IADNet). *Input:* observed particle trajectories (left). *IADNet module:* trajectories are mapped to node features, combined into edge features, and processed by a graph-structure-learning block whose weights $\mathcal{W}_0$ are initialized randomly. During training the network simultaneously minimizes a graph learning loss $\mathcal{L}_{GL}$ and a physics-based loss $\mathcal{L}_{phy}$. *Output:* the learned weight matrix is thresholded to yield a sparse interaction matrix $\mathcal{W}$ and its corresponding graph, which exposes the recovered particle couplings. Detailed definitions of feature construction, losses, and thresholding are provided in the *Methods* section.

supervised structure prediction [10–12]. It should be noted that some current research methodologies attempt to directly decouple the dynamical structure of systems through inverse inference from trajectory data [13,14]. While these frameworks effectively enhance the accuracy of dynamics prediction by modeling interaction patterns through the implicit edge-wise representations of GNNs, the generated implicit graph structures often remain entangled in latent feature spaces, hindering direct physical interpretation. Consequently, the mapping from learned correlations to the *true* interaction topology is often opaque.

To meet these challenges, we propose the **Interaction Discovery Neural Network (IADNet)**, a graph-learning architecture that reconstructs an explicit interaction matrix $\mathcal{W}$ directly from trajectory data, *without any a priori knowledge of connectivity*. As sketched in Fig. 2, IADNet outputs an interaction matrix $\mathcal{W}$ that characterizes particle interactions based solely on trajectory data, where the entries of the matrix encode the strength of the interaction and the symmetry of the matrix encodes the symmetry of the interaction potential. IADNet delivers three principal benefits:

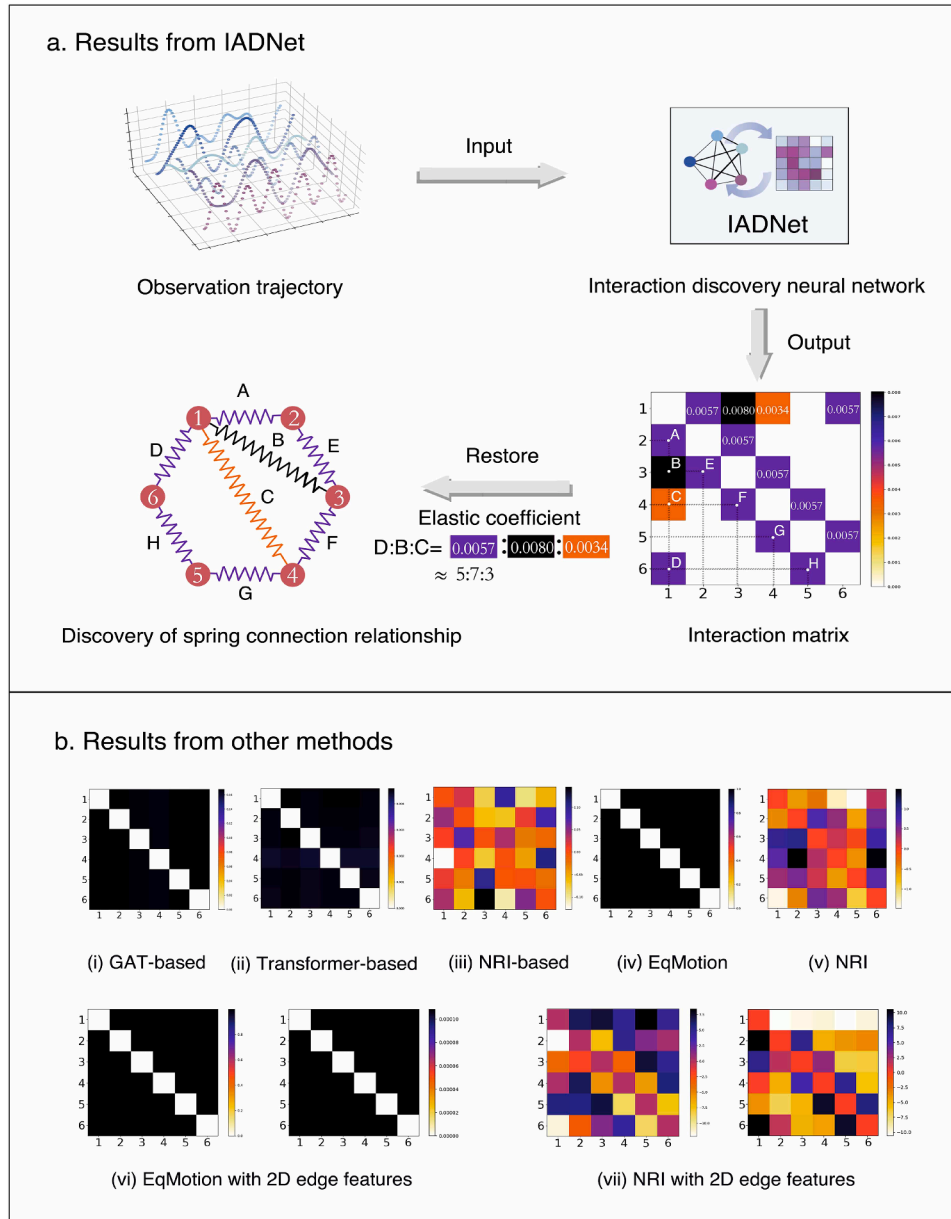


Fig. 3. Discover the interactions for N-ball-spring Systems. a. Observed trajectories (upper left) are fed to IADNet, which returns the interaction matrix shown on the right; rows and columns index the six balls and color encodes learned elastic strength (white = 0). The reconstructed graph (lower left) reproduces the ground-truth topology, including the two cross-links, and captures the elastic coefficients ratio 5:7:3 indicated beneath the matrix. The symmetry of the interaction matrix reflects the symmetric nature of the elastic potential energy. b. describes the interaction results obtained by inputting the same data into other network models. The feature dimension of the interaction in the first row is set to 1, and the feature dimension of the interaction in the second row is set to 2. Because these baselines learn time-dependent relations, we plot the temporal average of their edge weights. Only the interaction matrix generated by IADNet describes the true connections and properties of the system.

- The discovered interactions deepen our understanding of the system's characteristics, such as identifying defects in materials or uncovering molecular structures.
- The learned interaction topology provides a foundation for GNN-based methods, bypassing the need for prior structural knowledge, a limitation of conventional GNN approaches.
- The discovered interactions enable the application of graph theory to study multi-particle systems, providing new avenues for system analysis.

IADNet's efficacy has been validated across four distinct systems, where the input data consists solely of observed particle trajectories, and it is unknown whether any interactions exist:

- **N-ball-spring system (classical mechanics):** IADNet successfully identified spring connections between particles and estimated the corresponding elastic coefficients from the trajectory data.
- **Molecular dynamics (chemical system):** The method uncovered chemical bonds between atoms, effectively determining molecular structures from the trajectories.
- **Peyrard-Bishop-Dauxois DNA model (biological system):** IADNet discovered long-range interactions within the DNA structure and retrieved genetic information by analyzing molecular trajectories.
- **Fermi-Pasta-Ulam-Tsingou model (lattice system):** The model detected anomalies such as missing interactions and unexpected coupling behaviors purely from trajectory patterns, demonstrating robustness in anomaly detection.

The research results indicate that the revealed interactions are topologically equivalent to the description of interactions in physical systems and demonstrate the broad applicability and potential of IADNet in discovering particle interactions, offering a new tool for the study of multi-particle systems without relying on prior knowledge of interactions.

2. Methods

2.1. Problem formulation

We consider a system of N particles with unobservable interactions among them, whose strength is time-invariant; these interactions may be dense or sparse in the system. The recorded trajectories of these particles are represented by:

$$\mathcal{Z} = \{\mathbf{z}_i^t = (\mathbf{q}_i^t, \mathbf{p}_i^t) \mid 1 \leq i \leq N; t_1 \leq t \leq t_T\}, \quad (1)$$

where $\mathbf{q}_i^t \in \mathbb{R}^d$ and $\mathbf{p}_i^t \in \mathbb{R}^d$ denote the generalized coordinates and momenta of the i -th particle at time t , respectively. The index i represents the particles, and $t \in \{t_1, \dots, t_T\}$ refers to the time steps. Note that the observation intervals for the trajectory may be non-uniform, sparse (see Appendix).

Given the observed trajectory data $\mathcal{Z}$, the primary challenge is to develop an inference framework that can discover the interactions between arbitrary particle pairs (i, j) , assuming that the interactions are unknown a priori.

To formalize this problem, we model the system as a graph structure learning task. The system is represented by a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{W})$, where the vertex set $\mathcal{V} = \{v_i\}_{i=1}^N$ corresponds to the particles, each represented by a node. On the other hand, the edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents pairwise interactions between particles, which we seek to discover. The weight matrix $\mathcal{W}$ quantifies the interaction strengths between particles, where each element $w_{i,j}$ represents the strength of interaction between particles i and j , with the rows and columns of $\mathcal{W}$ corresponding to particles in $\mathcal{V}$.

The core task is to infer both the interaction graph $\mathcal{E}$ and the weight matrix $\mathcal{W}$ from the available observational data, without prior knowledge of the interaction structure. Specifically, this involves reconstructing the interaction topology (i.e., which particles interact) and estimating the interaction intensities (i.e., the strength of interactions) solely based on the observed trajectories of the particles. This challenge is schematically illustrated in Fig. 1.

2.2. Interaction discovery neural network (IADNet)

To address the challenge of interaction discovery in multi-particle systems, we introduce the Interaction Discovery Neural Network (IADNet), which is based on a graph structure learning strategy. IADNet takes the observed trajectory data $\mathcal{Z}$ (from Eq. (1)) as input and outputs an interaction matrix $\mathcal{W}$, which approximates the weighted adjacency matrix of the system. This matrix inherently encodes both the topological connectivity of the system (via sparsity patterns) and the interaction strengths (via the magnitude of the entries).

The structure of IADNet can be broken down into the following components:

Since the existence of interactions is initially unknown, we begin by constructing a randomly initialized graph $\mathcal{G}^0 = (\mathcal{V}, \mathcal{V} \times \mathcal{V}, \mathcal{W}_\theta)$, where the edge weights $\mathcal{W}_\theta$ are randomly initialized. The self-loops, e_{ii} , can optionally be retained depending on physical constraints identified during training.

The features of each node at time t are derived from the observed trajectories in phase space. Specifically, the node feature for particle i at time t is represented as:

$$\mathbf{h}_i^{(t)} = \mathbf{z}_i^t \in \mathbb{R}^{2d}, \quad \forall i \in \mathcal{V}, t \in \{t_1, \dots, t_T\}. \quad (2)$$

Since edges in the graph correspond to interactions between particles, and interactions are primarily governed by the relative motion of particles, we define the edge features as follows:

$$\mathbf{e}_{ij}^{(t)} = \begin{cases} [\mathbf{h}_j^{(t)} - \mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)} \odot \mathbf{h}_i^{(t)}], & \text{if } d = 1, \\ [\mathbf{h}_j^{(t)} - \mathbf{h}_i^{(t)}, (\mathbf{h}_j^{(t)} \odot \mathbf{h}_i^{(t)}) / (\|\mathbf{h}_j^{(t)}\| \cdot \|\mathbf{h}_i^{(t)}\|)], & \text{if } d > 1. \end{cases} \quad (3)$$

Here, $\odot$ denotes the Hadamard product. The first term captures the spatial distance between particles, while the second term encodes angular distance, based on cosine similarity.

Aggregating edge features over the graph allows us to produce a scalar output $\mathbf{G}^{(t)} \in \mathbb{R}$, which encodes system-level properties, such as potential energy or other interaction characteristics:

$$\mathbf{G}^{(t)} = \sum_{i \in \mathcal{V}} \sum_{j \in \mathcal{V}} \hat{w}_{i,j} g_{\text{encoder}}(\mathbf{e}_{ij}^{(t)}), \quad \forall i \in \mathcal{V}, t \in \{t_1, \dots, t_T\}, \quad (4)$$

where g_{encoder} is a encoder (e.g., a multi-layer perceptron) that evolves the edge features and maps them to a scalar value, while $\hat{w}_{i,j} \in \mathcal{W}_\theta$ represents the interaction strength between particles i and j .

We create a derivative dataset $\dot{\mathcal{Z}}$ (used as training labels) by introducing supplementary observation points separated by a small time interval h . The finite difference method is used to approximate the time derivatives:

$$\dot{\mathcal{Z}} = \{\dot{\mathbf{z}}_i^t = (\dot{\mathbf{q}}_i^t, \dot{\mathbf{p}}_i^t) \mid 1 \leq i \leq N; t_1 \leq t \leq t_T\}, \quad (5)$$

where $\dot{\mathbf{z}}_i^t = (\mathbf{z}_i^{t+h} - \mathbf{z}_i^t)/h$.

A multitask loss function is defined to integrate both graph learning and physical constraints:

$$\mathcal{L} = \frac{1}{T} \sum_{t=1}^T \mathcal{L}_{\text{phy}} + \gamma \mathcal{L}_{\text{GL}}, \quad (6)$$

where $\mathcal{L}_{\text{GL}} = \|\mathcal{W}_\theta\|_F^2$ is the graph learning loss, and $\mathcal{L}_{\text{phy}} = f(\mathbf{G}^{(t)})$ is the physical constraint loss. The hyperparameter $\gamma \in [0, 1]$ balances the two terms. If $\mathbf{G}^{(t)}$ represents the system's energy, the physical constraint loss is:

$$f(\mathbf{G}^{(t)}) = \left\| \frac{\partial \mathbf{G}^{(t)}}{\partial \mathbf{p}^t} - \dot{\mathbf{q}}^t \right\|_2^2 + \left\| -\frac{\partial \mathbf{G}^{(t)}}{\partial \mathbf{q}^t} - \dot{\mathbf{p}}^t \right\|_2^2. \quad (7)$$

Alternatively, if the force $\mathbf{F}$ of the system is observable, $\mathbf{G}^{(t)}$ can represent the potential energy, subject to:

$$f(\mathbf{G}^{(t)}) = \left\| \mathbf{F}^t + \frac{\partial \mathbf{G}^{(t)}}{\partial \mathbf{q}^t} \right\|_2^2, \quad (8)$$

where $\mathbf{F}^t$ is the supervised training label derived from the ground-truth forces at time t .

After training, the weight matrix $\mathcal{W}_\theta = (\hat{w}_{i,j})$ is obtained. A thresholding procedure is applied to simplify the matrix for interpretation and visualization:

$$w_{i,j} = \begin{cases} \hat{w}_{i,j}, & \text{if } |\hat{w}_{i,j}| > \lambda \text{ and } i \neq j, \\ 0, & \text{if } |\hat{w}_{i,j}| \leq \lambda \text{ and } i \neq j, \\ 0, & \text{if } i = j. \end{cases} \quad (9)$$

Here, λ is a threshold parameter, defined as:

$$\lambda = \kappa \mathcal{W}_{\max}, \quad \mathcal{W}_{\max} = \max_{0 \leq i, j \leq N, i \neq j} |\hat{w}_{i,j}|. \quad (10)$$

where κ is a hyperparameter. This thresholding process produces a simplified interaction matrix, facilitating both visualization and interpretation of the discovered particle interactions.

As a result, IADNet outputs an interaction matrix $\mathcal{W} = (w_{i,j}) \in \mathbb{R}^{N \times N}$, where N represents the number of particles in the system. The matrix elements $w_{i,j}$ for $i \neq j$ encode the strength of the interaction between particles i and j . When $w_{i,j} = 0$, it indicates no interaction between particles i and j . If $w_{i,j} = w_{j,i}$, it suggests that the interaction potential between particles i and j is symmetrical.

The general training pipeline of IADNet is presented in Algorithm 1. In our research, $\kappa = 0.35$.

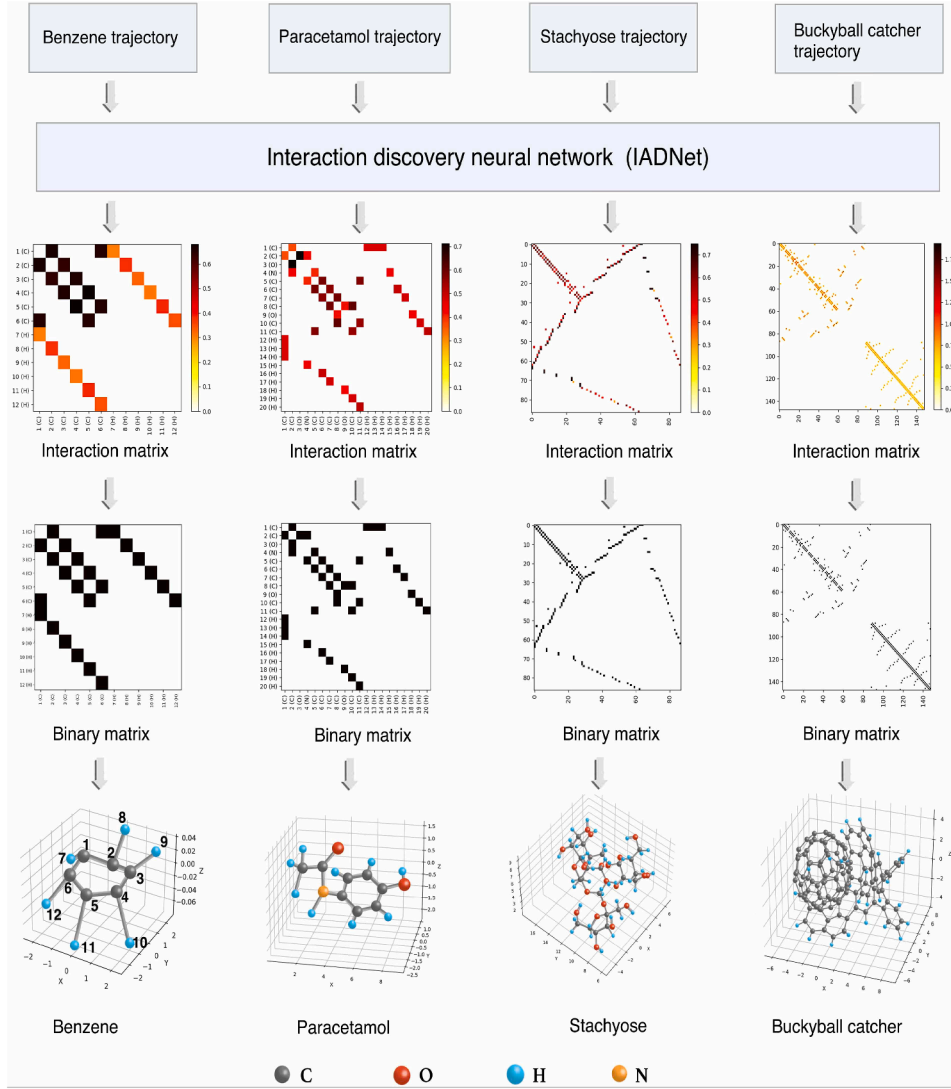


Fig. 4. Discover the chemical bond for molecules. IADNet generates an interaction matrix (visualized as a heatmap) based on molecular trajectories, followed by a binarization process. Entries with non-zero values are assigned a value of 1 (indicating chemical-bond presence), while entries equal to zero remain unchanged (indicating no chemical-bond). The molecular structure (shown in the last row) is reconstructed using the binary matrix. For example, the entry 1 between row 12 and column 6 in the binary matrix signifies a chemical bond between $H_{[12]}$ and $C_{[6]}$ in the molecular structure. The molecular structure reconstructed using this binary matrix is consistent with the classical valence bond theory.

3. Experiments

In this section, we selected four examples covering different disciplinary fields to verify the effectiveness of the method.

3.1. Discovering spring connection relationships in N -ball-spring system.

This subsection addresses the identification of spring connections in classical mechanical systems, specifically those composed of N balls interconnected by springs. In this dynamical system, the balls are connected by elastic springs that exhibit linear deformation under tension or compression. Each ball i is connected exclusively to its neighboring balls, $i - 1$ and $i + 1$, through springs, forming a closed-loop topology. This physical system has been widely used to evaluate the trajectory prediction capabilities of GNN-based models [15–18]. Before training the model, the system is abstracted into a graph structure where balls are treated as nodes, and spring interactions are mapped to edges. In this study, it is assumed that the spring connections cannot be directly

observed, which corresponds to the identification problem of implicit interaction relationships in practical engineering.

Moreover, we introduce two springs with distinct elastic coefficients to test the model's ability to distinguish between different springs. The system's dynamic behavior is described by the following Hamiltonian equations:

$$H = \sum_{i=1}^N \left\{ \frac{m_i}{2} \dot{x}_i^2 + \frac{k_{i,i-1}}{2} (x_i - x_{i-1})^2 \right\} + \frac{k_{1,3}}{2} (x_1 - x_3)^2 + \frac{k_{1,4}}{2} (x_1 - x_4)^2$$

where m_i represents the mass of ball i , $k_{i,j}$ denotes the elastic coefficient of the spring between balls i and j , and x represents the displacement of the balls. In our setup, we set $N = 6$, $m_i = 1$, $k_{i,i-1} = 0.5$, $k_{1,3} = 0.7$, and $k_{1,4} = 0.3$.

Fig. 3a demonstrates that IADNet recovers every genuine spring connection relationship between the balls and estimates the ratio of elastic coefficients between the springs. The symmetry of the interaction matrix reflects the quadratic potential of the system. This research achievement has demonstrated important application value in multiple fields, such as providing a data-driven method for modeling multi-particle system

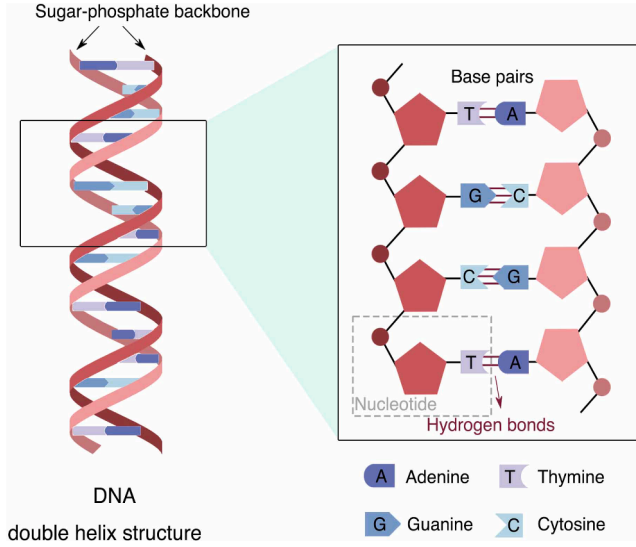


Fig. 5. The structure of the DNA double helix.

Algorithm 1 Calculate interaction matrix $\mathcal{W}$.

Input: Trajectories $\mathbf{z}_i^t \in \mathbb{R}^{2d}$

Output: Interaction matrix $\mathcal{W}$

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1: Initialization graph  $\mathcal{G}^0 = (\mathcal{V}, \mathcal{V} \times \mathcal{V}, \mathcal{W}_\theta)$  with random weight matrix
    $\mathcal{W}_\theta = (\hat{w}_{i,j})$ 
2: Node feature  $\mathbf{h}_i^{(t)} \leftarrow \mathbf{z}_i^t$ 
3: if  $d = 1$  then
4:   Edge feature  $\mathbf{e}_{ij}^{(t)} \leftarrow [\mathbf{h}_j^{(t)} - \mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)} \odot \mathbf{h}_i^{(t)}]$ 
5: else
6:   Edge feature  $\mathbf{e}_{ij}^{(t)} \leftarrow [\mathbf{h}_j^{(t)} - \mathbf{h}_i^{(t)}, (\mathbf{h}_j^{(t)} \odot \mathbf{h}_i^{(t)}) / (\|\mathbf{h}_j^{(t)}\| \cdot \|\mathbf{h}_i^{(t)}\|)]$ 
7: end if
8: for epoch  $\in [1, M]$  do
9:   for  $t \in [t_1, t_T]$  do
10:     $\mathbf{G}^{(t)} \leftarrow \sum_{i \in \mathcal{V}} \sum_{j \in \mathcal{V}} \hat{w}_{i,j} g_{\text{encoder}}(\mathbf{e}_{ij}^{(t)})$ 
11:     $\mathcal{L}_{\text{GL}} \leftarrow \|\mathcal{W}_\theta\|_F^2$ 
12:     $\mathcal{L}_{\text{phy}} = f(\mathbf{G}^{(t)})$ 
13:   end for
14:    $\mathcal{L} \leftarrow \frac{1}{T} \sum_{t=t_1}^{t_T} \mathcal{L}_{\text{phy}} + \gamma \mathcal{L}_{\text{GL}}$ 
15:   Adam optimization
16: end for
17:  $\mathcal{W}_{\text{max}} \leftarrow \max_{0 \leq i,j \leq N, i \neq j} |\hat{w}_{i,j}|$ 
18:  $\lambda \leftarrow \kappa \mathcal{W}_{\text{max}}$ 
19: if  $i = j$  then
20:    $w_{i,j} \leftarrow 0$ 
21: else
22:   if  $|\hat{w}_{i,j}| > \lambda$  then
23:      $w_{i,j} \leftarrow \hat{w}_{i,j}$ 
24:   else
25:      $w_{i,j} \leftarrow 0$ 
26:   end if
27: end if
28: return  $\mathcal{W} = (w_{i,j})$ 

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structures. The coefficients learned from the interaction matrix provide a promising method for equipment failure risk detection, manifested by relatively low strength coefficients at the risk of component fracture.

Fig. 3b benchmarks IADNet against interaction-learning baselines. Only substituting the definition of interaction matrix $\mathcal{W}$ in IADNet with a Graph Attention Neural Network (GAT) [19] (i), the Transformer architecture [20] (ii), or relational inference encoding module of Neural Relational Inference (NRI) [13] model (iii), or using Equivariant Multi-

agent Motion Prediction (EqMotion) [14] (iv and vi) or NRI (v and vii) directly, the visualization results of the interactive relationships they learn cannot reveal the real connections. The hyperparameter settings for each module follow the original configurations from references [19], [21] and [13], respectively. 3b (i), (ii), and (iii) verify the necessity of the interaction matrix definition method proposed in this paper. 3b (iv), (v), (vi) and (vii) illustrate that although existing methods may perform well in trajectory prediction accuracy, the learned interaction relationships mainly exist at the level of network hidden space features, and there is a lack of clear correspondence between interpretable physical topology structures. It should be noted that we did not replace $\mathcal{W}_\theta$ with EqMotion because it did not converge within our framework. Experimental results of Fig. 3b(i)-3b(iii) reveal that even with the application of physical constraints, the true interaction relationships within the system cannot be adequately captured without employing our defined matrix formulation.

3.2. Chemical-bond discovery in molecular trajectories

Molecules are formed by chemical bond connections between atoms. This intrinsic graph-like structure, where atoms serve as nodes and chemical bonds as edges, has spurred the development of graph neural network (GNN)-based methods for investigating molecular properties. Notable implementations of such methods include E(3)-equivariant deep learning (DeepH-E3) [22], which uses molecular Hamiltonians as both training labels and prediction targets for Hamiltonian matrix estimation. Additionally, frameworks such as the Molecular Hamiltonian Network (HamNet) [23], E(n)-Equivariant Graph Neural Networks (EGNN) [24], and Vector-Scalar Interactive Graph Neural Network (ViSNet) [25] leverage molecular properties as supervised labels to predict quantities such as energy states, interatomic forces, and dipole moments. Dual-View Molecular Representation Contrastive Learning (DVMR-CL) [26], and Adaptive Mixture GCN-Transformer (AdaMGT) [27] use molecular structure for molecular representation learning.

Here, we are faced with a more challenging scenario in which the presence of chemical bonds is unknown. We validate the effectiveness of IADNet using two challenging molecular dynamics benchmark datasets: the MD17 dataset [28] and the MD22 dataset [29]. Further details of these datasets can be found in the supplementary materials.

As shown in Fig. 4, the interaction matrix generated by the IADNet network accurately reconstructed the recognized configurations of various test molecules. It is worth noting that the model exhibits excellent scale adaptability - from benzene ring molecules containing only twelve atoms to buckyball catchers containing hundreds of atoms, the existence of chemical bonds can be accurately discovered.

This study emphasizes the discovery of chemical bond existence, an essential task with significant implications. First, the deterministic nature of chemical bonds offers a promising method for uncovering the topological structures of previously unknown molecules. Second, this analysis provides critical graph-structured inputs for GNN-based molecular modeling methods, laying the groundwork for future computational frameworks. More importantly, identifying functional groups via chemical bond existence allows further analysis of molecular properties. For example, IADNet identifies that paracetamol contains a benzene ring, an acetyl group, and a hydroxyl group.

It should be noted that this study only focuses on the determination of the existence of chemical bonds - confirming the bonding relationship between i and j when any entry $w_{ij} \in \mathcal{W}$ or $w_{ji} \in \mathcal{W}$ exhibits a non-zero value. To visualize clearly, we applied symmetry preprocessing to the interaction matrix.

3.3. Discovering hidden long-range couplings and sequence inference in an idealised DNA model

This section discusses whether IADNet can: (i) uncover long-range interactions in a DNA-like double helix and (ii) recover base-pair

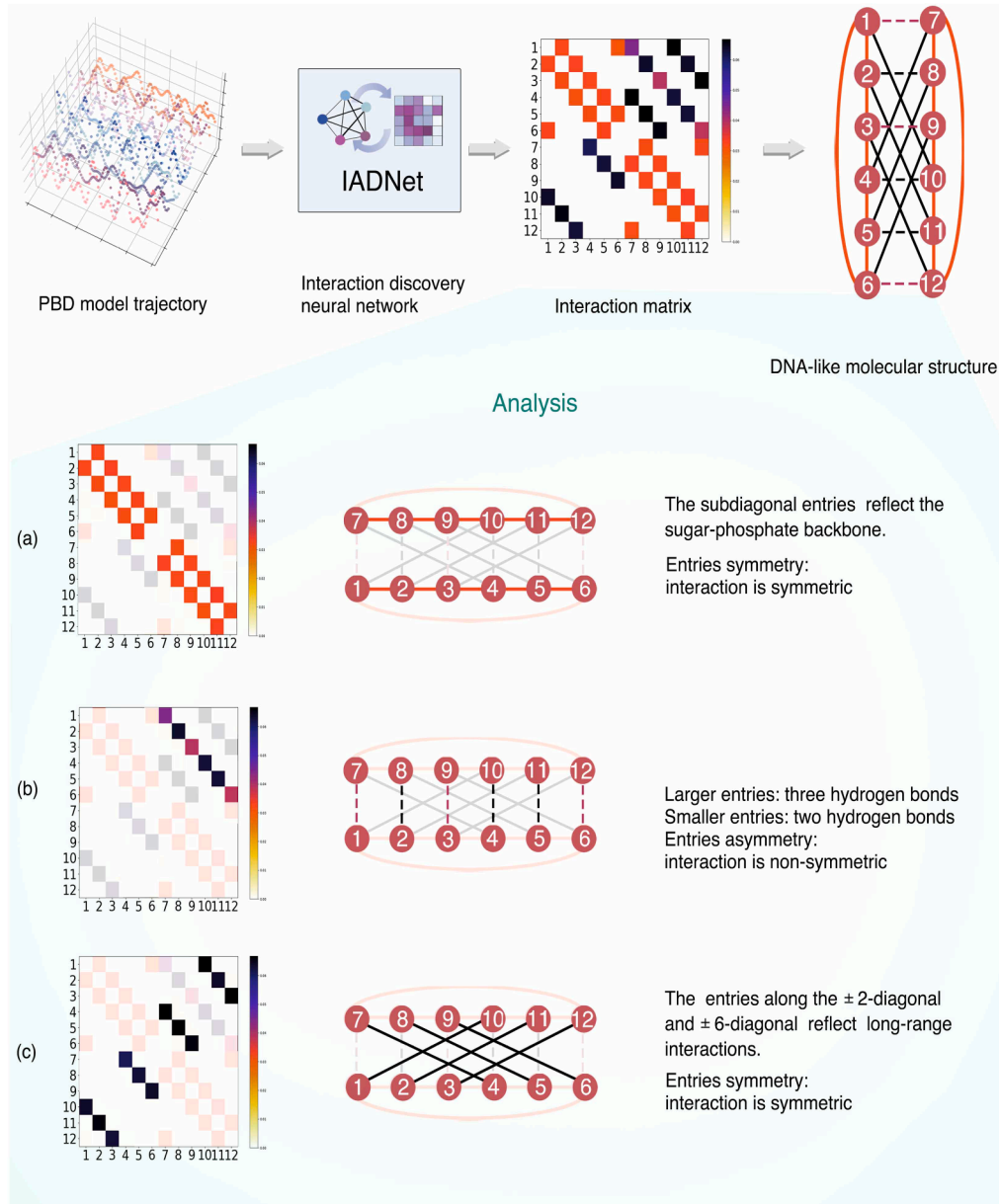


Fig. 6. Discover the DNA-like structure. Upon processing PBD model trajectories, IADNet generates an interaction matrix (heatmap) and a reconstructed DNA-like molecular structure. Dashed lines represent hydrogen bonds, with purple and black indicating two and three hydrogen bonds, respectively. The matrix row/column indices correspond to particle indices in the DNA-like structure. Symmetric matrix entries indicate symmetric interparticle potentials, while asymmetric entries reflect non-symmetric interactions. The decomposition of the matrix is shown on the left (panels (a)-(c)), with corresponding spatial configurations on the right (panels (a)-(c)).

ordering directly from dynamics. We employ an idealized DNA model for experimental validation, with the goal of demonstrating that our proposed methodology can systematically identify interaction patterns in systems exhibiting diverse interaction characteristics, similar to the DNA-like model.

The DNA molecule is a double helix structure formed by two antiparallel deoxyribonucleotide chains, which are connected through complementary base pairing (see Fig. 5). The backbone of each chain consists of alternating deoxyribose and phosphate groups, linked by 3',5'-phosphodiester bonds, forming the sugar-phosphate backbone. Specific pairing between the two strands results in base pairs, where nitrogenous bases strictly follow pairing rules: adenine (A) pairs with thymine (T), and cytosine (C) pairs with guanine (G). This selective pairing arises

from the spatial configuration and hydrogen-bonding capabilities of the bases: A-T pairs are connected by two hydrogen bonds, while C-G pairs are linked by three hydrogen bonds.

The Peyrard-Bishop (PB) model [30,31] was developed to simulate DNA transcription and has been successful in predicting melting temperatures and expanding melting curve shapes. This study uses its extended version, the Peyrard-Bishop-Dauxois (PBD) model [32,33], which incorporates the helical twist of DNA molecules. Specifically, the double helix structure implies that nucleotides from opposing strands can engage in long-range interactions mediated by water filaments when brought into proximity. This results in interactions between a nucleotide at position i on one strand and nucleotides at positions $(i \pm h)$ on the complementary strand. The dynamic behavior of the system is described by the

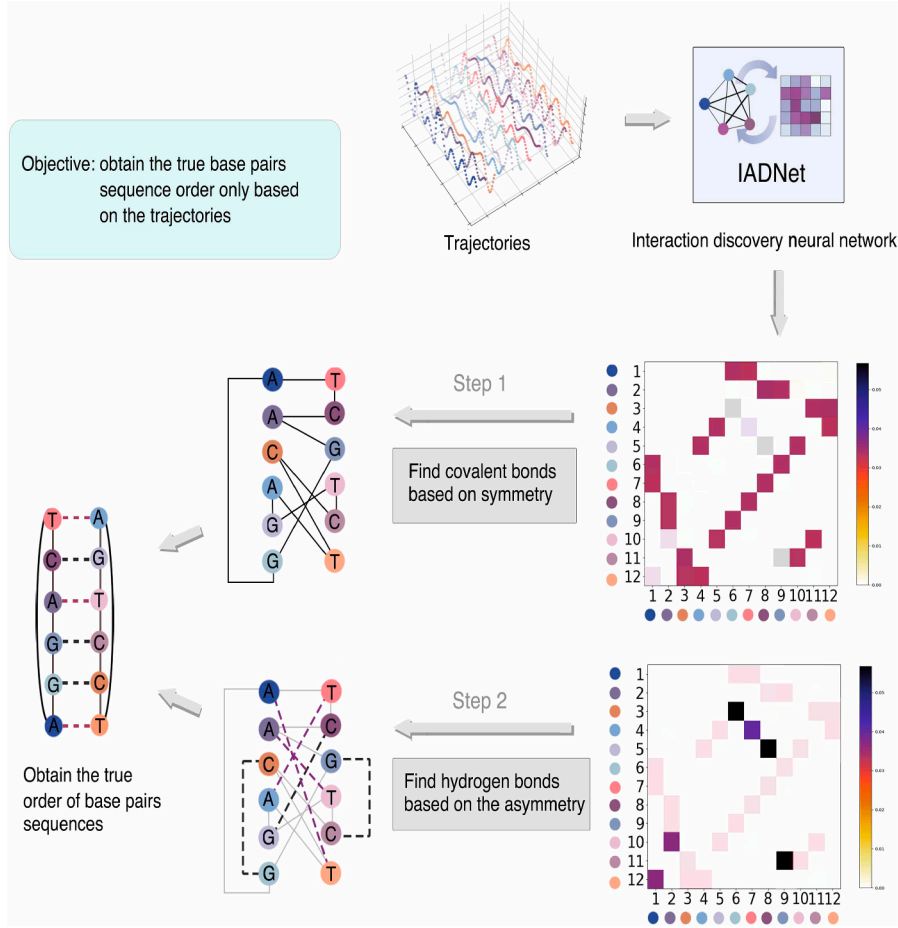


Fig. 7. Restore base pair sequence sorting. When the nucleotide sequence corresponding to the trajectory does not match the true base pair sequence order, IADNet can still correctly obtain the true base pairing arrangement. The trajectory colors correspond to the particle colors, with the row and column indices of the matrix matching the particle colors. The molecular structure diagram represents interactions between particles as lines. Dashed lines represent hydrogen bonds, with purple and black indicating two and three hydrogen bonds, respectively. Step 1 identifies adjacent relationships based on symmetric matrix components, while Step 2 identifies base-pairing relationships based on the asymmetric components. Finally, the correct base pair arrangement is reconstructed.

following Hamiltonian:

$$H = \sum_{i=1}^N \left\{ \frac{m}{2} (\dot{u}_i^2 + \dot{v}_i^2) + \frac{k}{2} [(u_i - u_{i-1})^2 + (v_i - v_{i-1})^2] \right\} + \sum_{i=1}^N \left\{ \frac{K}{2} [(u_i - v_{i+h})^2 + (u_i - v_{i-h})^2] + D_i [e^{-a(u_i - v_i)} - 1]^2 \right\}, \quad (12)$$

where k and K are the harmonic constants of the longitudinal and helical springs, respectively, u_i and v_i are the transversal displacements of the nucleotides, and D_i and a are the depth and inverse width of the Morse potential well, respectively. In our setup, we use $N = 6$, $m = K = k = a = 1$, and $h = 3$ for convenience. The boundary condition is chosen to be periodic. To reflect the difference in the number of hydrogen atoms between different base pairs, we set $D_i = 0.2$ for $i = 1, 3, 6$ and $D_i = 0.3$ for $i = 2, 4, 5$.

(i). Long-range interaction discovery: Fig. 6 shows that the interaction matrix generated by IADNet effectively captures the structure and interaction properties of DNA molecules with long-range interactions. Specifically, the asymmetric part of the interaction matrix coefficient reflects the asymmetry of hydrogen bond potential energy, while the symmetric part of the interaction matrix coefficient reflects the symmetry of interactions on the sugar-phosphate backbone and long-range interactions. By varying the amplitudes of matrix elements, the distinc-

tion between different numbers of hydrogen bonds has been achieved.

(ii). Sequence reconstruction: Genetic information is encoded in the sequence of DNA base pairs. Fig. 7 illustrates an additional experiment to validate whether IADNet can accurately determine the nucleobase sequence order based solely on the trajectories. This experiment adopts the zero long-range interaction assumption (i.e., $K = 0$). Consider a reference genome with the ground truth base-pair sequence TCAGGA, which corresponds to the complementary nucleobase sequence AGTCCT. Our experiment accounts for potential sampling artifacts: The observation device did not capture the original nucleotide sequence during sampling, leading to an incorrect order in the trajectory dataset. Specifically, the obtained trajectory data corresponds to the nucleotide sequence of AACAGGTCT, and the complementary relationship among them is also unknown. Upon feeding the obtained trajectories into IADNet, the generated interaction matrix can restore the true ordering of base sequences. The specific process is as follows: Step 1. Determine adjacent relationships by analyzing symmetric components of the interaction matrix. Step 2. Based on the exponential nature of the interaction potentials between base pairs, identify base-pairing partners by analyzing the asymmetric components of the interaction matrix. Finally, the true base pair arrangement is reconstructed. This experiment demonstrates the potential of IADNet for exploring particle ordering in systems where only trajectories are observable.

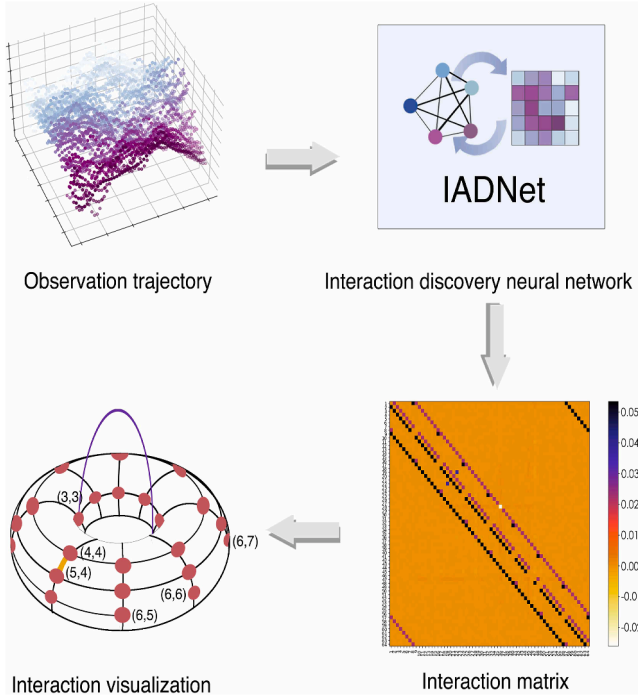


Fig. 8. Anomaly detection in lattice. Using the system's observed trajectories as input, the interaction matrix (heatmap) generated by IADNet successfully reveals anomalous phenomena in the lattice system. The row and column indices of the matrix correspond to particle indices in the system diagram, where connecting lines in the diagram denote the presence of interactions. In the heatmap color scheme, yellow regions map to zero values, signifying the absence of coupling among particles. $w_{22,19} = w_{19,22}$ indicates a symmetric long-range interaction between particles 22 in position (3,6) and 19 in position (3,3), while $w_{45,46} = w_{46,45} = w_{47,46} = w_{46,47} = 0$ shows no interaction among particles 45 in position (5,6), 46 in position (6,6), and 47 in position (6,7). Asymmetric interactions are captured as $w_{36,28} \neq w_{28,36}$, with non-zero values revealing a non-symmetric interaction between particles 28 in position (4,4) and 36 in position (5,4). The remaining interactions follow nearest-neighbor lattice connectivity in a square grid. Boundary conditions $|w_{i,i+56}| > 0$ and $|w_{1+8(i-1),8i}| > 0$ reflect the boundary conditions $x_{1,j} = x_{8,j}$ and $x_{i,1} = x_{i,8}$, respectively.

3.4. Anomaly detection in lattice system

The Fermi-Pasta-Ulam-Tsingou (FPUT) problem serves as a foundational example in the study of statistical mechanics, known for uncovering complex dynamical behaviors and its central role in the development of nonlinear dynamics research [34]. Decades after its inception, the FPUT problem continues to stimulate active research, with many of its scientific implications still under exploration [35]. This study focuses on the two-dimensional version of the FPU model [36,37]. Before this physical system can be used as a benchmark for evaluating GNN-based models in particle trajectory prediction tasks, it is essential to abstract the system into a graph-structured input, where particles are represented as nodes and their interactions as edges [38].

The central assumption of this work is that the ground-truth particle interaction relationships are unknown. To emulate manufacturing defects, we introduce three local perturbations:

- (i) the particle at (6, 6) is disconnected from its two horizontal neighbors;
- (ii) a square potential interaction is inserted between particles (3, 3) and (3, 6), creating a long-range coupling;
- (iii) the interaction potential between (4, 4) and (5, 4) is changed from square cubic mixture to cubic.

Therefore, the governing Hamiltonian for the system is expressed as follows:

$$H = \sum_{i=1}^N \sum_{j=1}^M \left(\frac{\dot{x}_{i,j}^2}{2} + V(x_{i+1,j} - x_{i,j}) + V(x_{i,j+1} - x_{i,j}) \right) - V(x_{6,7} - x_{6,6}) - V(x_{6,6} - x_{6,5}) + \frac{1}{2}(x_{3,3} - x_{3,6})^2 - \left(\frac{1}{2}(x_{5,4} - x_{4,4})^2 - \frac{1}{15}(x_{5,4} - x_{4,4})^3 \right), \quad (13)$$

where the potential function $V(u)$ is defined as:

$$V(u) = \frac{1}{2}u^2 + \alpha \frac{1}{3}u^3,$$

with α representing the strength of the nonlinear interaction potential. For the purpose of our experimental setup, we set $M = N = 8$ and $\alpha = 0.25$. The boundary conditions are defined as: $x_{1,j} = x_{8,j}$ for $j = 1, \dots, 8$, and $x_{i,1} = x_{i,8}$ for $i = 1, \dots, 8$.

Fig. 8 demonstrates the interaction reconstruction effects of IADNet based on the system's observed trajectory. The interaction matrix performs a row-wise reindexing of particles in an 8×8 lattice system. Specifically, the matrix indices are assigned such that index 1 corresponds to the particle at grid position (1, 1), index 2 to the particle at (1, 2), and so on, with index 64 corresponding to the particle at grid position (8, 8). The analysis of the interaction matrix not only reflects the interactions between particles forming a grid and the existence of interactions on both sides of the grid due to boundary conditions, but also shows that most of the interactions between particles are asymmetric, with one asymmetric interaction having a different strength than the others. In addition, two missing interactions and one symmetric long-range interaction were found. The results indicate that IADNet has potential applications for defect detection in lattice systems.

The appendix comprehensively documents methodological details, including: (1) dataset introduction with network configuration and training specifications; (2) comparative analysis between IADNet-generated weight matrix $\mathcal{W}_\theta$ and threshold-processed interaction matrix $\mathcal{W}$; (3) noise robustness experiments and extended testing on MD17/MD22 molecular systems; (4) ablation studies examining edge feature construction and graph learning loss formulations; (5) systematic investigations into how observation integration time-step sizes and time window settings impact model performance; (6) chaotic system interaction identification.

4. Conclusion

This paper presents IADNet, a neural network framework developed to discover interparticle interactions directly from observed particle trajectories, even if the existence of interactions is unknown a priori. The effectiveness of IADNet has been validated across four distinct systems from diverse scientific domains, each presenting unique challenges. The results demonstrate both the breadth and the interpretability of the approach.

In the N-ball spring-mass system (classical mechanics), IADNet successfully identified spring connections between particles and accurately determined the relative elastic coefficients, relying solely on observed trajectories. This demonstrates the model's capability to uncover implicit interactions, especially in engineering contexts where direct observation of interactions is often not feasible.

For molecular dynamics, tests on the MD17 and MD22 datasets (chemistry) demonstrated that IADNet can recover chemical bonding information from molecular trajectories, enabling the determination of molecular structures. These results suggest that the model offers a promising approach for structure determination based solely on trajectory data, which could have significant implications in molecular studies.

In the context of DNA dynamics (biology), IADNet uncovered long-range interactions within the DNA double-helix structure and successfully derived the base-pair sequencing from these interactions. This

showcases IADNet's ability to identify interaction patterns in systems with varied interaction modes, such as DNA-like models, and to accurately determine the spatial arrangement of particles.

When applied to the Fermi-Pasta-Ulam-Tsingou (FPUT) system (lattice), IADNet not only accurately mapped the interaction relationships but also detected anomalies, such as missing interactions and unexpected coupling behaviors. These findings indicate the model's potential as a tool for anomaly detection in dynamic systems.

Beyond these case studies, the interaction matrices yielded by IADNet serve as ready-made inputs for downstream graph neural networks and standard graph-theoretic analyzes, providing a bridge between raw trajectory data and higher-level modeling.

IADNet is currently applicable to systems with static structural parameters during the observation window. When structural coefficients (referring to elasticity coefficient, gravity coefficient, etc) vary dynamically, the model requires retraining based on new trajectory data or extension of the framework to time-dependent graphs. Similarly, the current energy-based loss objective preserves conservative dynamics; this constraint could be mitigated by introducing trainable dissipation terms or incorporating auxiliary networks specifically designed to simulate non-conservative forces. Future research priorities include developing dynamic topology adaptation mechanisms and implementing non-conservative variants of the framework. Despite these limitations, IADNet's unique capability to generate interpretable interaction graphs—in contrast to opaque latent representations—positions it as a promising tool for diverse applications in physical sciences, particularly in structural health monitoring, risk assessment frameworks, and data-driven discovery processes.

CRediT authorship contribution statement

Ru Geng: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis; **Yixian Gao:** Writing – review & editing, Supervision, Methodology, Funding acquisition; **Jian Zu:** Writing – review & editing, Validation, Supervision, Methodology; **Hong-Kun Zhang:** Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition.

Data availability

MD17, MD22 are available in ref. [28] and ref. [29], respectively.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Dataset

For the N-ball spring system, FPUT model, and PBD model, their trajectory data is generated using the explicit symplectic Runge-Kutta-Nyström algorithms (Section 8.5.3 in [39]), with a time step of 0.001s over a 30-second simulation period. The initial conditions are set as follows:

For 1D configurations:

$$\begin{aligned} q_i^0 &= \lambda_i \sin\left(\frac{(i-1)\pi}{N-1}\right), \\ p_i^0 &= 0, \quad i = 1, \dots, N, \end{aligned} \quad (\text{A.1})$$

For 2D configurations:

$$\begin{aligned} q_{i,j}^0 &= \lambda_{i,j} \sin\left(\frac{(M(i-1) + (j-1))\pi}{MN-1}\right), \\ p_{i,j}^0 &= 0, \quad i, j = 1, \dots, N. \end{aligned} \quad (\text{A.2})$$

with λ_i and $\lambda_{i,j}$ drawn from a standard normal distribution. The generated trajectories were sampled at 0.2-second intervals to construct the primary dataset, comprising 50 independent trajectories. Subsequent derivative datasets were constructed by temporal shifting of sampling points with 0.001-second offsets for differential estimation.

The introduction to the molecule dynamics dataset MD17 [28] and the molecule dynamics dataset MD22 [29] are systematically summarized in Table A.1. These datasets record molecular energy, atomic force, atomic position, and some atomic properties.

Table A.1
Introduction to molecular dynamics datasets.

Type	Name	Atoms number	Dataset samples
Aromatic hydrocarbon	Benzene	12	627,983
Aromatic compounds	Toluene	15	442,790
Fatty aldehyde	Malonaldehyde	9	993,237
Organic acid	Salicylic acid	16	320,231
Organic acid	Aspirin	21	211,762
Alcoholic compounds	Ethanol	9	555,092
RAN basic group	Uracil	12	133,770
Polycyclic aromatic hydrocarbons	Naphthalene	18	326,250
Acylation aniline	Paracetamol	20	106,490
Aromatic amine	Azobenzene	24	99,999
Lipids	DHA (docosahexaenoic acid)	56	69,753
Nucleic acids	AT-AT(DNA base pairs)	60	20,001
Carbohydrates	Stachyose	87	27,272
Supramolecules	Buckyball catcher	148	6,102

Appendix B. Training settings

B.1. N-Ball spring system, FPUT model and PBD model experiment

For the N-ball spring system, FPUT model, and PBD model, the encoder $g_{encoder}$ employs a three-layer fully-connected structure with 100 hidden units per layer, using the SiLU activation function. The physics-constrained loss function is defined by (7). The hyperparameter γ in (6) is 0.05. Furthermore, to enhance the model's robustness against data errors (noise) when increasing the numerical integration step size, we correspondingly raise the γ value. Specifically: at an integration step size of 0.01, γ is 0.1; At a step size of 0.1, γ is 0.5. The training parameters are configured as follows: The total number of epochs is set to 2000, with the Adam optimizer employed for parameter updates and a batch size of 256. A piecewise constant decay strategy [40] is adopted for learning rate scheduling: The initial learning rate is set to 10^{-3} , which decays to 10^{-4} at 1000 epochs, and further adjusts to 10^{-5} after 1500 epochs.

B.2. Molecular experiment

In the molecular experiment section, We set the hyperparameter γ to 0.005. The encoder $g_{encoder}$ adopts a three-layer fully-connected structure with a hidden layer dimension of 256. To ensure model robustness, each molecule underwent 10 independent training runs, and the final results were derived from the average of the 10 weight matrices. The core task of this study is chemical bond determination: if any element $|\hat{w}_{i,j}|$ or $|\hat{w}_{j,i}|$ in the weight matrix exceeds a predefined threshold, a chemical bond is identified between atoms i and j . To optimize data visualization, we symmetrized the weight matrix using: $\tilde{\mathcal{W}}_\theta = \frac{1}{20} \sum_{i=1}^{10} (\text{abs}(\mathcal{W}_\theta^i) + \text{abs}((\mathcal{W}_\theta^i)^T))$, where $\mathcal{W}_\theta^i$ denotes the weight matrix obtained from the i -th training. The symmetrized average matrix $\tilde{\mathcal{W}}_\theta$ was thresholded to generate the final molecular interaction matrix.

In terms of data partitioning, For standard experiments, we used 10,000 training samples and 2000 validation samples (for early stopping), with an early stopping patience of 20 epochs. For the buckyball catcher system (total 6102 samples), the dataset was adjusted to 5000 training and 1102 validation samples. The batch size was 256, but reduced to 32 for large molecular systems (buckyball, stachyose, AT-AT, and DHA) due to memory constraints. All models were trained using the Adam optimizer with a learning rate of 0.001 for 5000 iterations.

The threshold setting strategy is as follows: Except for aspirin, the threshold was uniformly set to 35% of the maximum absolute value in the average interaction matrix. For aspirin, an initial 35% threshold introduced isolated atom artifacts, which were eliminated by lowering the threshold to 25% of the maximum absolute value. Bond identification results for other molecules will be discussed in subsequent sections.

It should be specifically noted that this molecular dataset lacks velocity information. Nevertheless, given that complete atomic force data within the system are explicitly recorded in the dataset, we employ (8) to define the physical constraint term L_{phy} . Trajectory data $\mathbf{q}_i = (x_i, y_i, z_i)$ being the Cartesian coordinates of the atom i . This design choice of physical constraint aligns with standard methodologies in the field: Some GNN-based methods typically accomplish force prediction tasks by integrating molecular structural features, molecular trajectories, and force label data [22–25]. Our framework introduces no additional data modalities, especially from the perspective of providing molecular structures for these GNN-based methods.

Appendix C. The comparison of the weight matrix and the interaction matrix.

Fig. C.9 shows the weight matrix $\mathcal{W}_\theta$ (with imposed constraint $\hat{w}_{i,i} = 0$ for clearer visualization comparison) obtained after IADNet training and the interaction matrix obtained $\mathcal{W}$ based on our defined threshold. It can be seen from the figure that the weight matrix displays the main

features of the interaction, while the threshold we defined makes the interrelationships more clearly displayed.

Appendix D. More experiments

D.1. Adding noise

To verify the robustness of the model, we added Gaussian noise with a standard deviation of $\sigma^2 = 0.05$ on each data point in the training set. Fig. D.10 shows the comparison of the interaction matrix between IADNet input trajectory with and without 5% noise. It can be seen that they are basically the same, indicating the robustness of IADNet.

D.2. Effects of observation integration time-step sizes on model performance

In this section, we investigate the impact of numerical integration time-step size on the performance of IADNet. Here, the error introduced by the integration time-step size is treated as a form of noise accompanying the observed trajectories. Fig. D.11 displays the interaction matrices output by IADNet under time-step sizes of 0.001, 0.01, and 0.1. The results demonstrate that IADNet accurately identifies inter-particle interactions across different systems in all three cases. Moreover, the model correctly captures both the relative interaction strengths and the symmetry of the interaction potentials, except for the six-ball spring system with a time-step size of 0.1. These findings indicate that IADNet exhibits considerable robustness to integration step size variations.

D.3. Molecular chemical bond discovery

Fig. D.12 shows the chemical bond discovery results of ethanol, uracil, naphthalene, toluene, azobenzene, malonaldehyde, salicylic acid, aspirin, AT-AT, and DHA in the MD17 and MD22 datasets. The reconstructed molecular structure is consistent with classical valence bond theory.

D.4. Ablation experiment

In this section, we demonstrate the importance of each step in our framework.

• **Definition of $\mathbf{e}_{ij}^t$:** Fig. D.13 systematically compares the effects of different relative displacement definitions (including conventional approaches that only consider relative displacements) on toluene molecular training outcomes, thereby demonstrating the necessity of our proposed $\mathbf{e}_{ij}^t$ formulation. Specifically:

Fig. D.13 (a) shows the structure of the molecule, and (b) visualizes the corresponding adjacency matrix $\mathbf{A}$ with row and column labels consistent with the atomic labels in (a). Black pixel elements denote bonded connections (value=1), directly reflecting chemical bonding information between atoms; (c) presents the average weight matrix calculated using our proposed Eq. (6). Obviously, by selecting the threshold, we can obtain (b) from (c).

Fig. D.13 (d) is the average weight matrix obtained when

$$\mathbf{e}_{ij}^t = \left[\frac{\mathbf{h}_j^{(t)} \odot \mathbf{h}_i^{(t)}}{\|\mathbf{h}_j^{(t)}\| \cdot \|\mathbf{h}_i^{(t)}\|} \right], \quad i, j \in \mathcal{V}, \quad t = t_1, \dots, t_T. \quad (\text{D.1})$$

It is difficult to obtain (b) from (d) by setting a threshold, mainly because the weight coefficients of some H-C atomic pairs are significantly lower in matrix $\tilde{\mathcal{W}}_\theta$. The values of typical elements such as $\tilde{\mathcal{W}}_\theta(12, 4)$ are similar to the background color, making it difficult to distinguish whether it is training noise.

Fig. D.13 (e) is the average weight matrix obtained when

$$\mathbf{e}_{ij}^t = \left[\frac{\mathbf{h}_j^{(t)} - \mathbf{h}_i^{(t)}}{\|\mathbf{h}_j^{(t)} - \mathbf{h}_i^{(t)}\|} \right], \quad i, j \in \mathcal{V}, \quad t = t_1, \dots, t_T. \quad (\text{D.2})$$

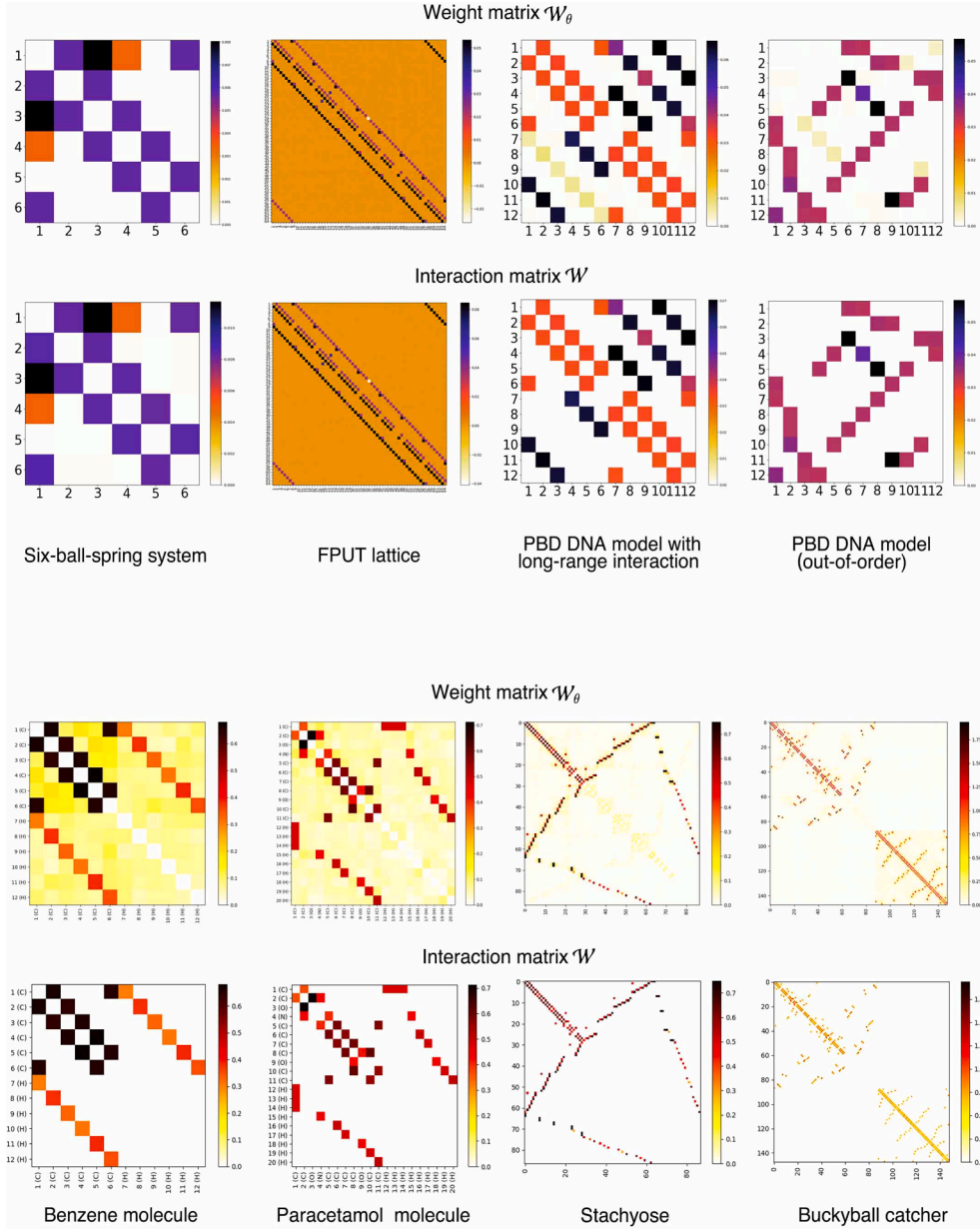


Fig. C.9. The comparison of the weight matrix $\mathcal{W}_\theta$ and the interaction matrix $\mathcal{W}$.

After selecting the threshold, (e) will identify chemical bonds that do not exist in toluene. For example, the weight between the 13th row and 2nd column of the average interaction matrix is extremely high, indicating the existence of a chemical bond between $H_{[13]}$ and $C_{[2]}$. Fig. D.13 (f) also identifies excess chemical bonds. Fig. D.13 (g) is the average weight matrix obtained when

$$\mathbf{e}_{ij}^t = \left[\mathbf{h}_j^{(t)} - \mathbf{h}_i^{(t)} \right], \quad i, j \in \mathcal{V}, \quad t = t_1, \dots, t_T, \quad (\text{D.3})$$

which also identifies redundant chemical bonds. For example, while recognizing the chemical bond between $C_{[11]}$ and $C_{[2]}$, it also recognizes the chemical bond between $C_{[4]}$ and $C_{[2]}$, as well as between $C_{[7]}$ and $C_{[5]}$.

• **Definition of graph learning loss $\mathcal{L}_{GL}$:** If $\mathcal{L}_{GL}$ is omitted, the model cannot effectively learn interaction effects. Still taking (11) as an example, Fig. D.14 (a) demonstrates the interaction learning performance when γ is set to zero. The results show that the model fails to establish the correct spring connections, indicating the critical importance of this term for proper interaction learning. Furthermore, $\mathcal{L}_{GL}$ cannot be

set to the L_1 norm as it imposes sparsity constraints, but the interactions among particles in the system may not necessarily be sparse. Fig. D.14 (b) shows the heatmap of the interaction matrix output by the model using L_1 norm, which contains limited information and cannot effectively reflect the even symmetry of potential energy like $\mathcal{L}_{GL}$.

D.5. Effects of observation window settings on model performance

This section investigates the impact of trajectory observation time windows on model performance. We selected the 2D FPUT, DNA, and spring systems as test cases for validation. Previously, we systematically studied the model performance of these systems when the observation time intervals (timesteps) were uniform. In this section, we focus on investigating the impact of non-uniformly distributed observation timestamps (random sampling) within different time windows on the model. To meet various practical needs, we design three distinct observation scenarios:

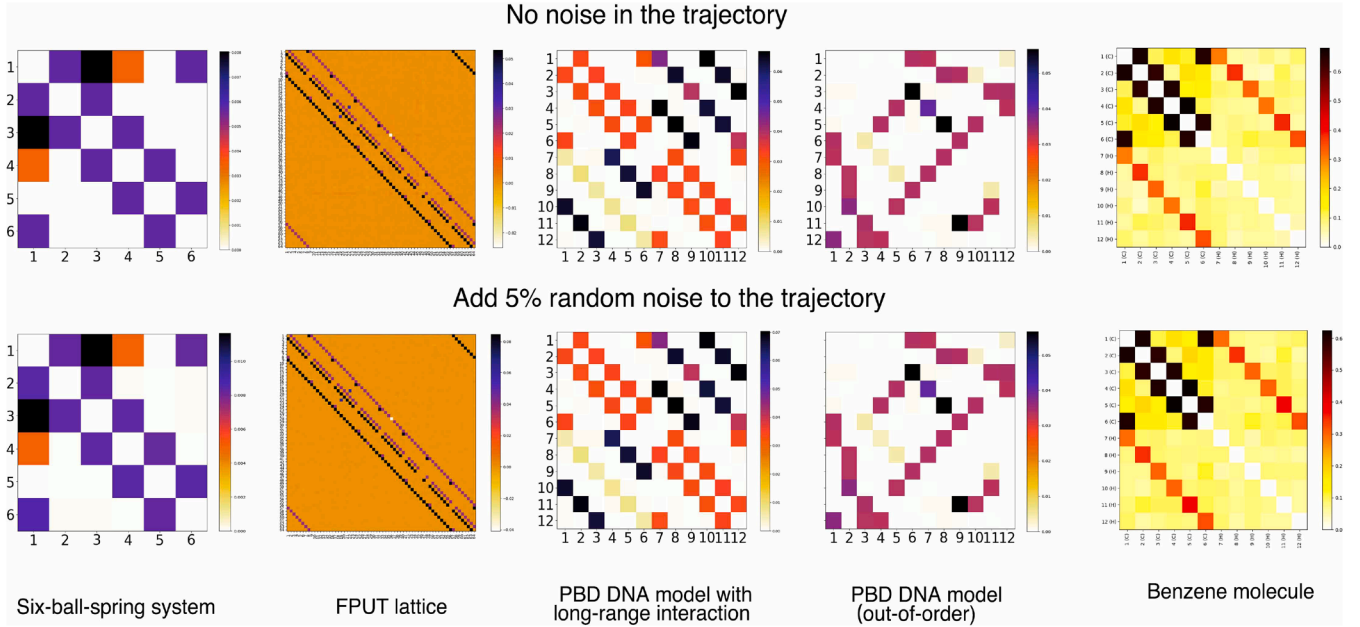


Fig. D.10. The comparison of the interaction matrix between IADNet input trajectory with and without 5% noise shows that the two are basically the same, indicating the robustness of IADNet.

(a). Multi-trajectory dense sampling: 50 trajectories with different initial conditions, each is randomly sampled 100 points within a 30s time window (total samples: 5,000);

(b). Multi-trajectory sparse sampling: 50 trajectories with different initial conditions, each is randomly sampled 5 points within a 5s time window (total samples: 250);

(c). Long-term single-trajectory sparse sampling: A single trajectory is randomly sampled 200 points over a 4,000s duration (total samples: 200).

Scenario (a) is used for comparison with the previous uniform sampling. Scenario (b) reflects the use of multiple trajectories observation when the observation time window is insufficient to describe system characteristics, and sparse observation is used to reduce costs, while scenario (c) demonstrates the use of sparse observation points to reduce costs when observing a single trajectory for a long time.

Fig. D.15 presents the IADNet reconstruction results for three dynamical systems under these observation conditions, with columns (a)-(c) corresponding to the three scenarios. It can be seen that across all test scenarios, IADNet accurately identifies the system's interaction topology. Notably, although the absolute coefficients in the system's interaction matrix may exhibit variability, the relative interaction strengths (coefficient ratio) among nodes show consistency. The only deviation occurs in the PBD system under scenario (c), where interaction strength estimation shows minor errors—attributable to the extremely challenging training conditions (merely 200 samples). This can be effectively mitigated by moderately increasing the number of samples.

In conclusion, the performance of IADNet is not sensitive to the choice of observation time window or sampling interval. The only prerequisite is having enough training samples. For the neural network model, this number is a hyperparameter. In all examples discussed in this paper, using ten thousand training samples proved to be sufficient for achieving robust performance.

D.6. Interaction identification in chaotic systems

This section investigates a FPUT chain system exhibiting chaotic behavior. Its governing equation is

$$H = \sum_{i=0}^N \left[\frac{p_i^2}{2} + \frac{(q_{i+1} - q_i)^2}{2} + \frac{1}{4} \frac{(q_{i+1} - q_i)^3}{3} \right] \quad (\text{D.4})$$

with fixed boundary conditions $q_0 = q_{N+1} = 0$ and $N = 7$.

We observed a trajectory with an energy of 1. Its Poincaré section (defined by the plane $p_5 = 0$, displayed on the $q_4 - p_4$ phase plane) over the time interval $[0, 50000\text{s}]$ is shown in Fig. D.16 (a), and its largest Lyapunov exponent is 0.042589. These characteristics indicate that the trajectory exhibits chaotic behavior. Sampling points from this trajectory are selected as training data. To investigate the impact of different recording time intervals on model performance, four distinct sampling strategies are adopted: randomly selecting 10,000 sample points in $[0, 4000\text{s}]$, randomly selecting 2000 sample points in $[0, 500\text{s}]$, equal interval sampling 4000 sample points in $[0, 4000\text{s}]$, and randomly selecting 5000 points in $[3500\text{s}, 4000\text{s}]$ for training. The interaction matrices learned from these four sampling schemes are presented in Fig. D.16 (b) - (e), respectively. The system interactions inferred from the interaction matrix are all presented in Fig. D.16 (f). The results indicate that all four sampling methods consistently identified the chain-like lattice structure of the FPUT system. The asymmetry of the interaction matrix suggests the presence of a non-even symmetric potential in the system. Although the coefficients in the interaction matrix vary, it is noteworthy that the ratios between the non-zero entries within the same matrix remain consistent. Specifically, the constant ratio among non-zero entries observed in this experiment indicates uniform interaction strength between different particles, which is consistent with the assumptions of the theoretical model.

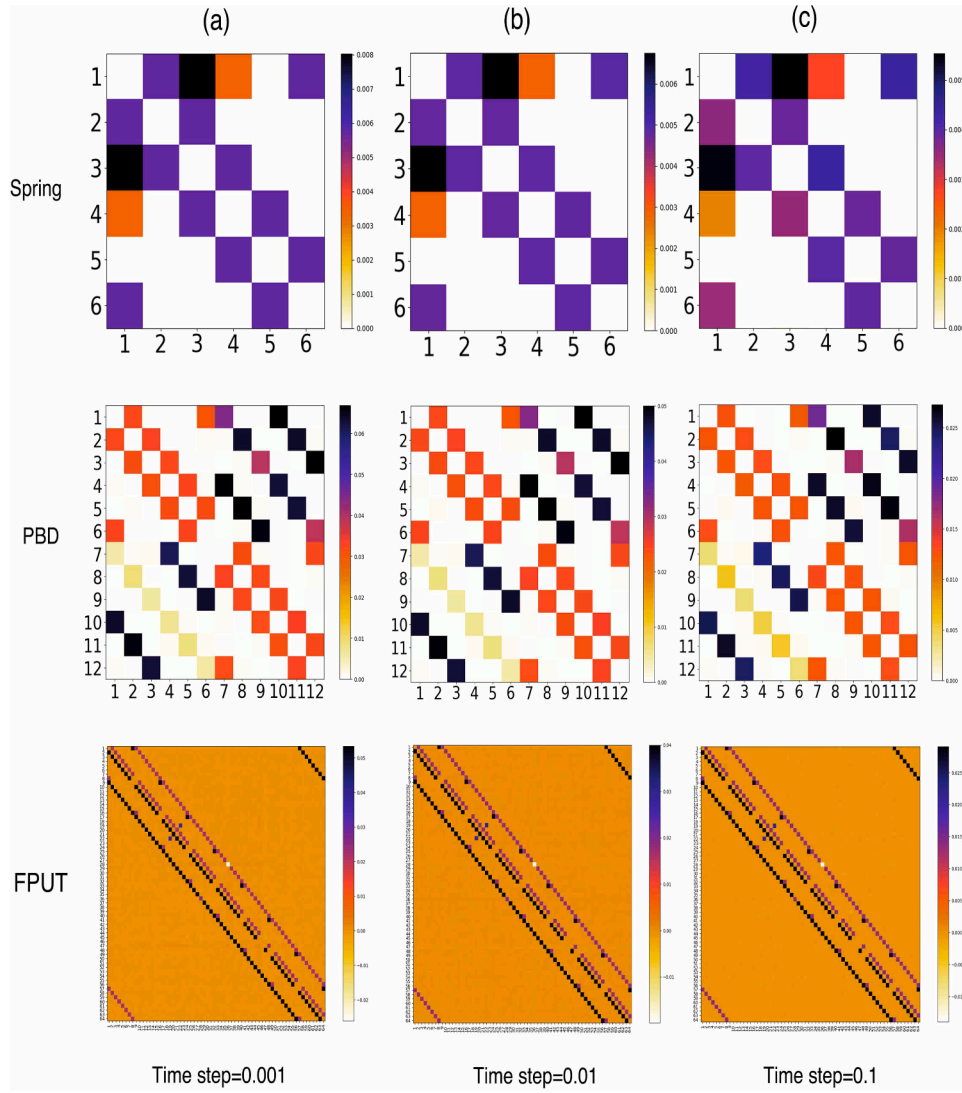


Fig. D.11. Comparison of interaction matrices output by IADNet using trajectories generated with different numerical integration time-step sizes.

The primary rationale for the prolonged observation was to ensure the training dataset captured a broader spectrum of chaotic dynamics, thereby providing a rigorous test of the model's performance on chaotic trajectories.

These findings demonstrate that IADNet is also effective in identifying interactions within chaotic systems, with no strict requirements on the observation time window or observation interval.

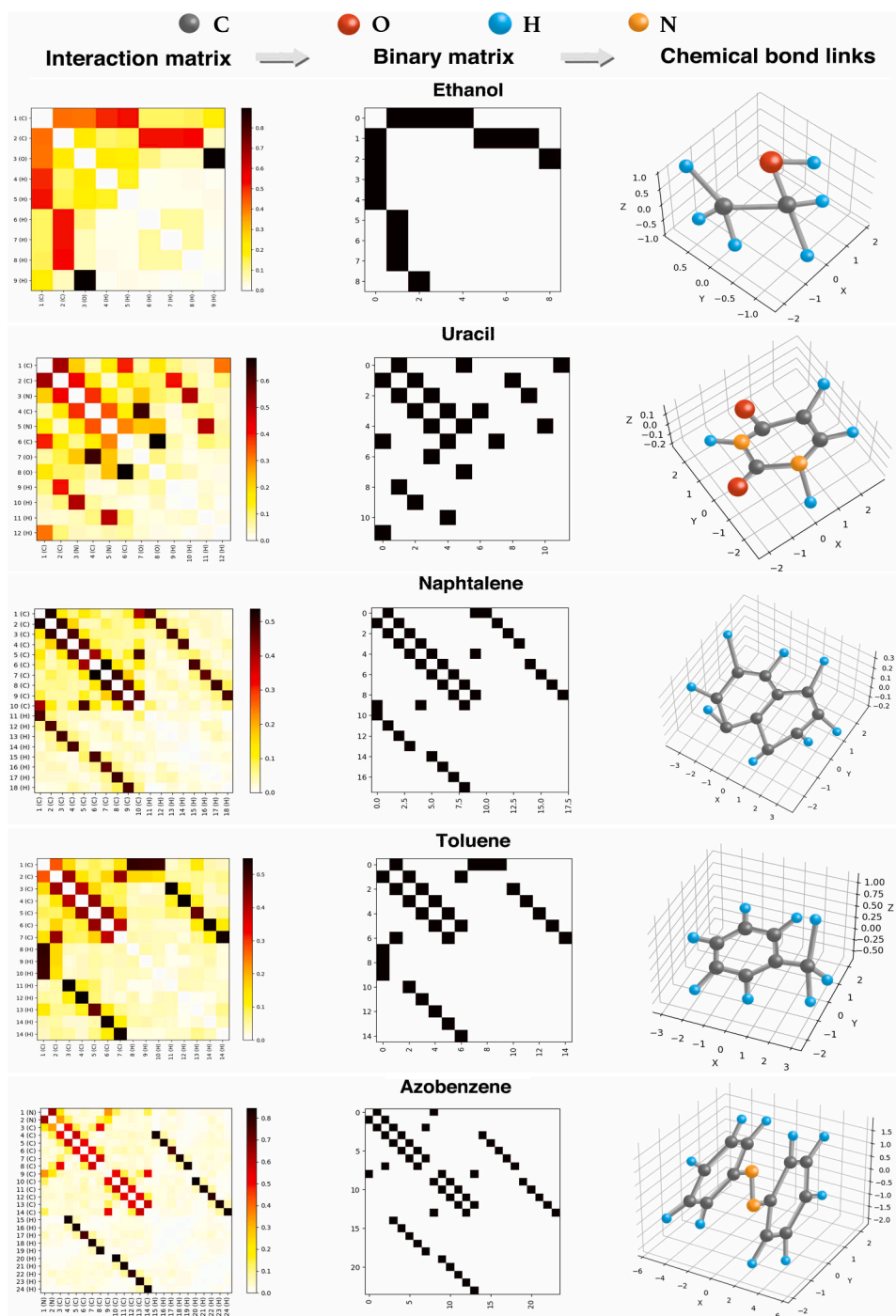


Fig. D.12. The three columns in the figure represent the average weight matrix over 10 training runs, the binary matrix obtained by thresholding, and the chemical bond connections derived from the binary matrix, respectively. The black regions in the binary matrix denote a value of 1, indicating the presence of a chemical bond between the atoms corresponding to the rows and columns of the binary matrix.

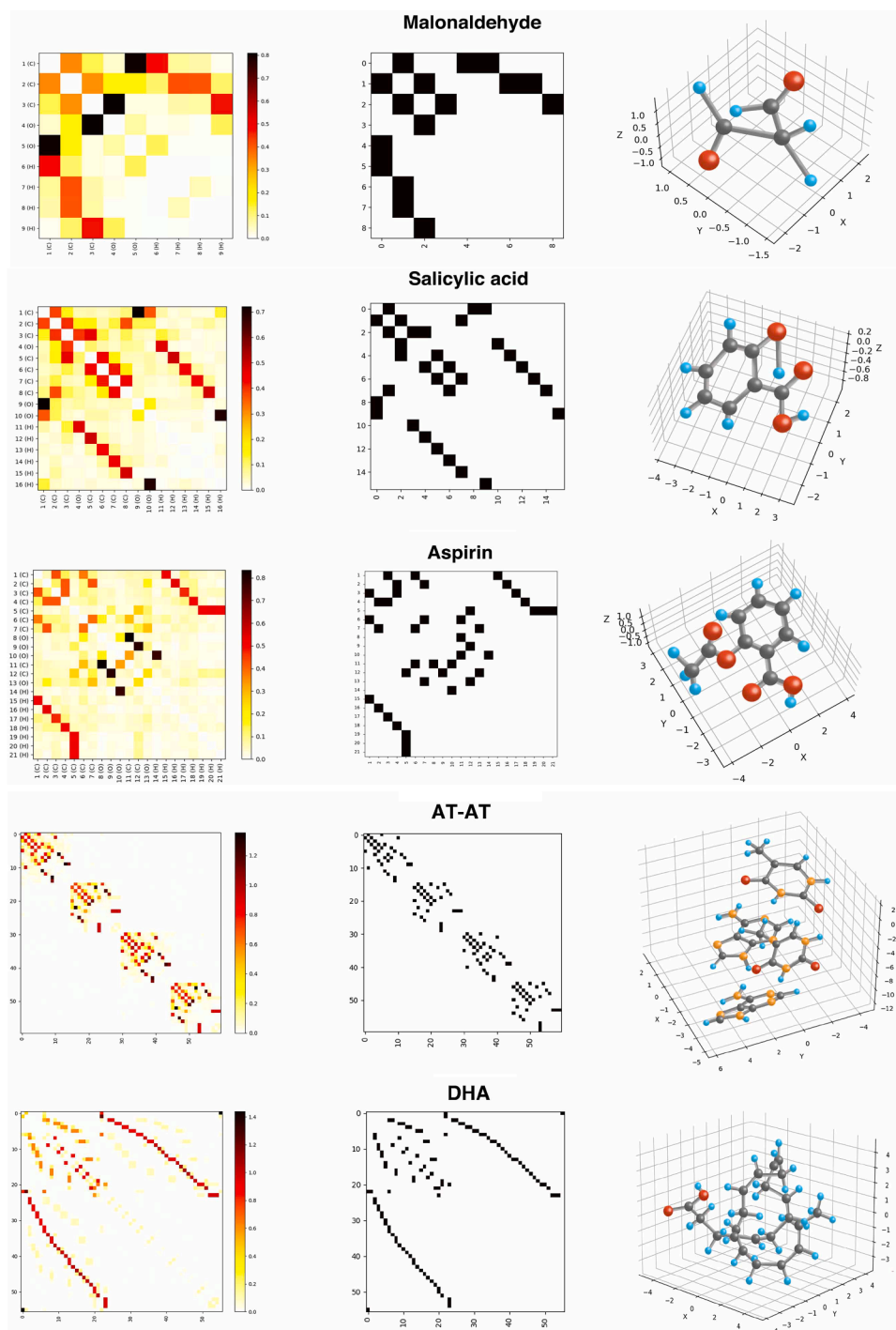


Fig. D.12. continue.

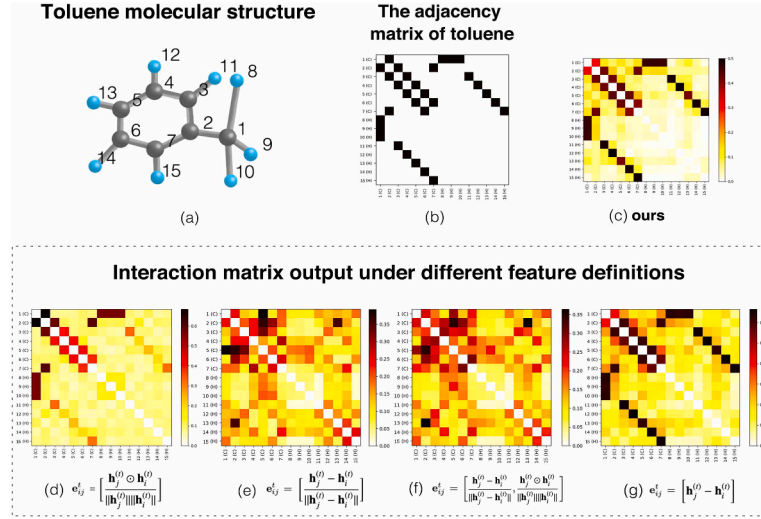


Fig. D.13. (a) is the molecular structure of the toluene, and (b) is the adjacency matrix A of the toluene. (c)-(g) are average weight matrixes $\tilde{\mathcal{W}}_p$ obtained by different definition of e_{ij}^t to train toluene. (b) can be obtained by taking the threshold of (c). From (d), it difficult to identify the chemical bond between $H_{[12]}$ and $C_{[4]}$. (e) and (f) indicate the existence of a chemical bond between $H_{[13]}$ and $C_{[2]}$. (g) indicates the existence of a chemical bond between $C_{[7]}$ and $C_{[5]}$.

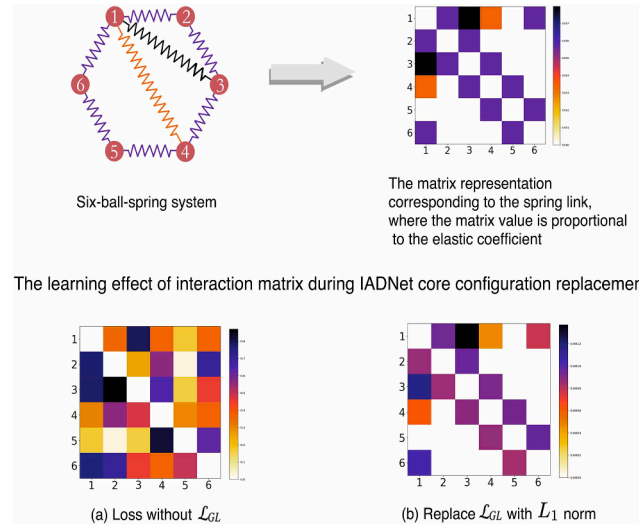


Fig. D.14. Different regularization loss learning effects for the six-ball-spring system described in (11).

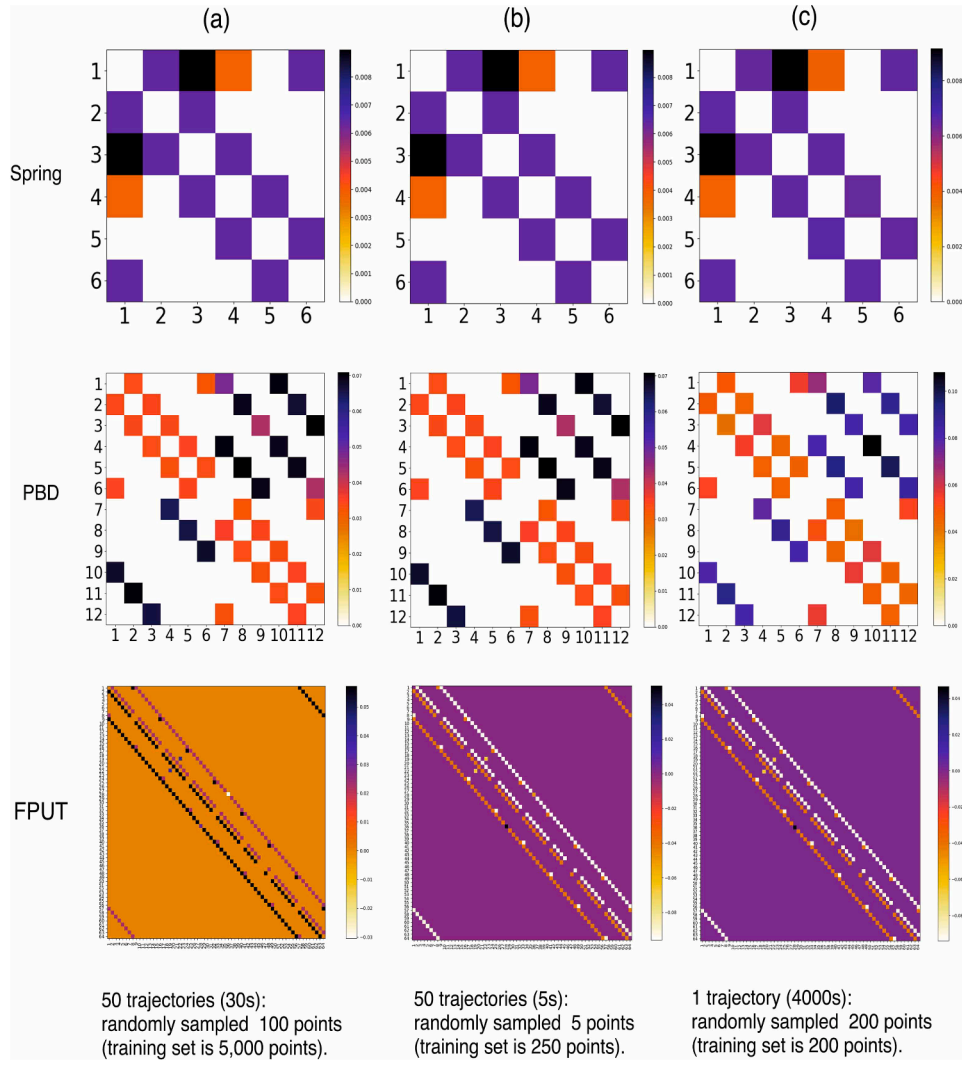


Fig. D.15. Comparison of interaction matrices generated by IADNet across different trajectory observation scenarios.

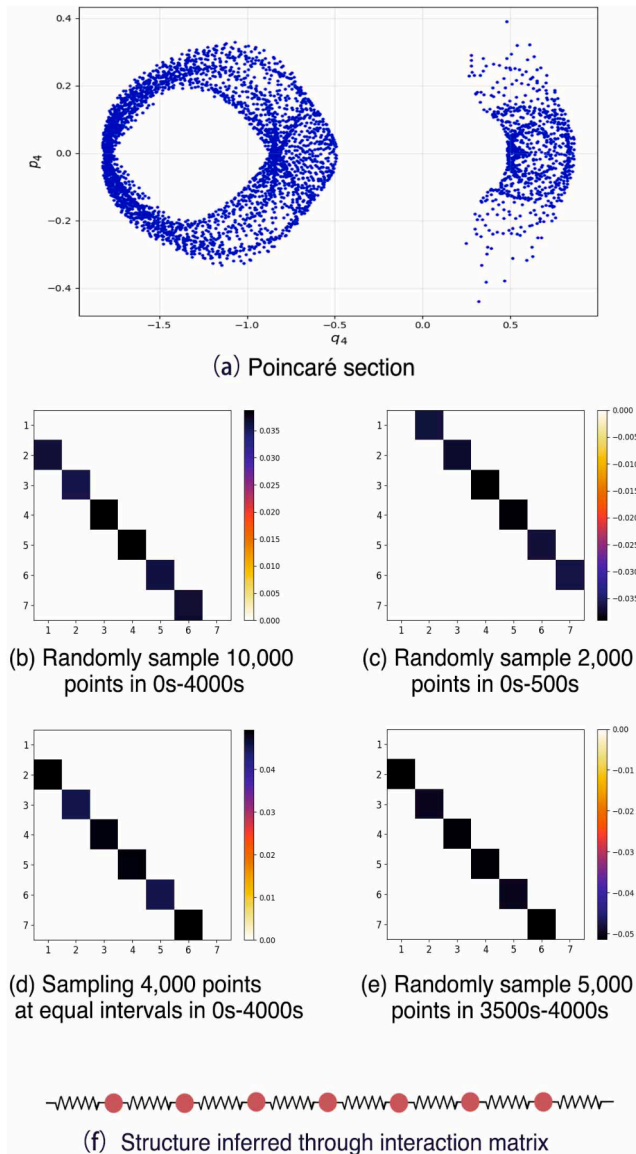


Fig. D.16. (a). Poincaré section (defined by the plane $p_5 = 0$, displayed on the $q_4 - p_4$ phase plane) derived from an observed trajectory of the FPUT lattice. (c)-(e). show the results of the interaction matrices trained on datasets constructed using four different trajectory sampling methods, respectively. (f) is the system interactions inferred from the interaction matrix.

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